

Date: 10.04.21

Group Study

Chapter: 05

IPV4 Address

5.1

What is IP Address? The identifier used in the 1st layer of the TCP/IP protocol suite to identify each device connected to the Internet is called the Internet address or IP Address.

An IPv4 Address is a 32-bit address that ~~are~~ unique and defines the connection of each host or a router to the internet and universally.

IPv4 Address are unique. That means each address defines one and only one connection to the internet.

Two devices on the internet can never have the same address at the same time.

The IPv4 Address are unique and universal.

☐ What is address space?

An address space is the total number of addresses used by the protocol.

If a protocol uses b bits to define an address, the address space is 2^b because each bit can have two different values (0 and 1).

☐ The address space of IPv4 protocol is 2^{32} (more than 4 billion)

☐ There are three common notation to show an IPv4 address :

- 1) Binary Notation (Base-2)
- 2) Dotted-decimal Notation (Base-256)
- 3) Hexa-decimal Notation (Base-16)

☐ In Binary Notation, an IPv4 address is displayed as 32-bit.

☐ An IPv4 address is referred to as a 32-bit address, a 4-octet address, or 4-byte address

Ex: 0110101 10010101 00011101 11101010

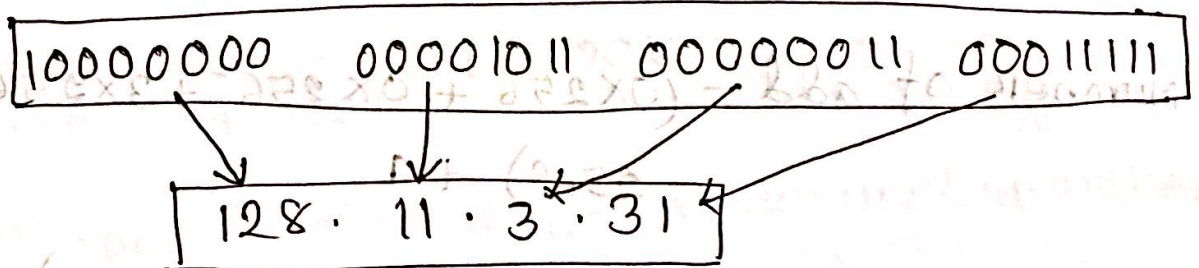
To make IPv4 address more compact and easier to read an IPv4 address is usually written in decimal form with a decimal point (dot) separating the bytes.

This format is referred to as dotted-decimal notation.

Each number in dotted-decimal notation is between 0-255.

Binary Notation

Ex:



Dotted-Decimal Notation

Hexadecimal notation:

- ⇒ Each Hexa-decimal digit is equivalent to 4 bits
- ⇒ a 32-bit address has 8 hexadecimal digits
- ⇒ Hexadecimal notation is often used in network programming.

Find the number of addresses in a range if the first address is 146.102.29.0 and the last address is 146.102.32.255

Solⁿ: Last add. - First Add.

146 . 102 . 32 . 255

146 . 102 . 29 . 0

0 . 0 . 3 . 255

Now, convert this to base 10 (or decimal) + 1

$$\therefore \text{Number of add} = (0 \times 256^3 + 0 \times 256^2 + 3 \times 256^1 + 255 \times 256^0) + 1$$

$$= (0 + 0 + 768 + 255 + 1)$$

$$= 768 + 256$$

$$= 1024$$

Ans

Example: 5.6

The first address in a range of address is

14.11.45.96. If the number of addresses in the range is 32, what is the last address?

Soln:

We convert the number of addresses to base 256 and minus 1 to base 256, which is $0.0.0.31$

We then add it to the first address to get the last address. Addition is in base 256.

$$\begin{aligned}\text{last address} &= (14.11.45.96 + 0.0.0.0.31)_{256} \\ &= 14.11.45.127 \quad (\text{Ans})\end{aligned}$$

Ex: 5.7

NOT

Apply ~~NOT~~ operation on a 32-bit binary Number:

$$\begin{array}{cccc}00010001 & 01111001 & 00001110 & 00100011 \\ \Rightarrow 11101110 & 10000110 & 11110001 & 11011100\end{array}$$

Apply NOT operation on a dotted-decimal number:

$$\begin{array}{cccc}17 & \cdot & 121 & \cdot & 14 & \cdot & 35 \\ \Rightarrow 238 & \cdot & 134 & \cdot & 241 & \cdot & 220\end{array}$$

[minus from 255]

Ex: 5.8 Bitwise AND operation:

- ① If 0 or 255 $\rightarrow$ select the smallest one.
- ② If either not 0 or 255 then $\rightarrow$ write each byte as a sum of eight terms, where each term is power of 2. Now, select the smaller term and add them to get the result.

First Number: 17 · 121 · 14 · 35

Second Number: 255 · 255 · 140 · 0

Result: 17 · 121 · 12 · 0

Powers: 128 64 32 16 8 4 2 1
2⁷ 2⁶ 2⁵ 2⁴ 2³ 2² 2¹ 2⁰

Byte(14): 0 0 0 0 8 4 2 0

Byte(140): 128 0 0 0 8 4 0 0

Result: 0 0 0 0 8 4 0 0

$$= 8 + 4$$

$$= 12$$

$$(14 \text{ AND } 140) = 12$$

Ex-5.9

Bitwise OR operation

First Number: 17 · 121 · 14 · 35

Second Number: 255 · 255 · 140 · 0

Result: 255 · 255 · 142 · 35

Powers: 128 64 32 16 8 4 2 1
2⁷ 2⁶ 2⁵ 2⁴ 2³ 2² 2¹ 2⁰

Byte(14): 0 0 0 0 8 4 2 0

Byte(140): 128 0 0 0 8 4 0 0

Result: 128 0 0 0 8 4 2 0

$$= 128 + 8 + 4 + 2$$

$$= 142$$

$$\begin{array}{r} 128 \\ 14 \\ \hline 142 \end{array}$$

5.2

CLASSFUL ADDRESS

In classful addressing, the ip address space is divided into five class: A, B, C, D and E.

Each class occupies some part of the whole address space.

class A : $2^{31} = 50\%$ address

class B : $2^{30} = 25\%$ address

class C : $2^{29} = 12.5\%$ "

class D : $2^{28} = 6.25\%$ "

class E : $2^{28} = 6.25\%$ "

In classful addressing, the address space is divided into five classes: A, B, C, D, and E.

Recognizing classes:

In binary notation, the first few bits can immediately tell us the class of the address.

In dotted-decimal notation, the value of the first byte can give the class of the address.

class A : 0.... [first bit of first octet is zero]

Ex: 01100010 10111100 00011110 11101100

class B : 10.... [first two bit of first octet is 10]

Ex: 101111100 00110011 10001110 01010111

class C: 110.... [first three bit of first octet is 110]

class D: 1110.... [first four bits of first octet is 1110]

class E: 1111.... [first four bits of first octet is 1111]



Class A: 0-127 [first byte]

Class B: 128-191

Class C: 192-223

Class D: 224-~~239~~ 239

Class E: 240-255

Dotted-Decim
Notation

———— X ————

Netid & Hostid

☐ In classful addressing, the an IP Address in classes A, B and C is divided into netid and hostid.

☐ classes D and E are not divided into netid or hostid.

class A : first byte = netid ; next three bytes =

hostid
class B : first two bytes = netid ; last two bytes
= hostid

class C : first three bytes = netid ; last one byte
= hostid.

class D : multicast address.

class E : Reserved for future use

Classes & Blocks

One problem with classful addressing is that each class is divided into a fixed number of blocks and with each block having a fixed size.

class A : $2^7 = 128$ blocks

total address per block = 16,777,216

Many address wasted in this class.

⊗
Class B : 2 bytes = $2 \times 8 = 16$ bits (16 - 2 fixed bits = 14)

Class B is divided into $2^{14} = 16,384$ blocks

Each block contains 65,536 addresses.

Class C :

Class C is divided into $2^{21} = 2,097,152$ blocks

Each block contains 256 addresses.

Class D :

There are just one block in class D addresses.

It is designed for multicasting.

Class E :

There is just one block of class E addresses. It was designed for use as reserved address.

Two Layer-Addressing:

Each address in classful addressing contains two parts:

i) netid

ii) hostid

The netid defines the network.

The hostid defines the particular host connected to that network.

If n -bits in the class defines the net, then $32-n$ bits defines the host.

Netid for:

class A : 8 bits

class B : 16 bits

class C : 24 bits

We normally like to know three pieces of information about a block:

i) The number of addresses

ii) The first address and

iii) The last address

▣ The number of addresses in a block, N , can be found using: $N = 2^{32-n}$

▣ To find the first address, we keep the n leftmost bits and set the $(32-n)$ rightmost bits all to 0s.

▣ To find the last address, we keep the n leftmost bits and set $(32-n)$ rightmost bits all to 1s.

Example 5.13

An address in a block is given 73.22.17.25.
Find the number of addresses in the block, the first address, and the last address.

Solⁿ:

73 is between (0-127), so it belongs to class A.

1) The value of n for class A = 8.

$\therefore$ Number of address in this block = 2^{32-8}
 $= 2^{24}$

2) To find first address we keep first 8 bits and set the rightmost 24 bits all to 0s.

∴ The first address in the block is $8 \cdot 730.0.0/8$
here, 8 is the value of n .

② The first address is called the network address and is not assigned to any host. It is used to define the network.

③ To find the last address we keep the first 8 bits and set the rightmost 24 bits all to 255.

∴ The last address of this block is

$73.255.255.255/8$.

* The last address is used for special purpose.

Ex-5.14

An address in a block is given as $180.8.17.9$. Find the number of address in the block, the first address and the last address.

Soln :

since, 180 is between 128 and 191, the class of the address is B.

1) The number of address in this block : $N = 2^{32-16}$
 $= 2^{16}$
 $= 65,536$

First address: $= 180.8.0.0$

Last address $= 180.8.255.255$

128	0	32	16	8
	2	1		
		255	164	
			= 132	
				127
				128

Ex-5.15

An address in a block is given $200.11.8.4$.

Find the number of address in the block
first address and last address.

Soln:

Since 200 is between ~~192-223~~ 192-223,
the class of the address is C.

1) Total address in this block, $N = 2^{32-24}$

$$= 2^8$$
$$= 256$$

2) First address $= 200.11.8.0/24$

3) Last address $= 200.11.8.255$

▣ The first address, network address, is particularly important because it is used in routing a packet to its destination network.

▣ The network address is the identifier of a network.

Network Mask

▣ A network mask or default mask in classful addressing is a 32-bit number with n leftmost bits all set to 1s and $(32-n)$ rightmost bits all set to 0s.

▣ We have 3 default mask in classful addressing:

1) Class A : ~~255.0.0.0~~ 255.0.0.0

2) Class B : 255.255.0.0

3) Class C : 255.255.255.0

▣ To extract the network address from the destination address of a packet, a router uses the AND operation.

Three Level Addressing

☐ We need more than two hierarchical levels, for two reasons -

- 1) Dividing large networks into several sub-networks for better security and management
- 2) Dividing large blocks into smaller sub-blocks to share ^{them} with other organizations.

☐ What is subnetting?

⇒ Subnetting is the practice of dividing a large network into smaller networks.

☐ The network mask is used when the network is not subnetted.

☐ When we divide a network to several subnetworks, we need to create a subnetwork mask (or subnet mask) for each subnetworks.

☐ Subnetting increases the length of netid and decrease the length of hostid.

☐	$n_{\text{sub}} = n + \log_2 s$	$n = \text{length of netid}$
		$n_{\text{sub}} = \text{length of each subnetid}$
		$s = \text{number of subnets, that must be power of } 2$

Subnet Address

Ex-5.21

Find Subnet Address:

Given address :	141	14	120	77	} <u>AND</u>
Subnet Mask :	255	255	192	0	
Subnet Address :	141	14	64	0	

120 $\rightarrow$ 0 + 64 + 32 + 16 + 8 + 0 + 0 + 0 $\otimes$
 192 $\rightarrow$ 128 + 64 + 0 + 0 + 0 + 0 + 0 + 0

64

- * 255 অথবা 0 থাকলে ছোট্ট হোটে যেটা হবে। যেমন 141, 255 থাকলে 141 হবে।
- * 255 অথবা 0 কোনটা না থাকলে Positional values এর AND হবে। যেহেতু শুধুমাত্র হাতের পড়িলে মিলে দ্বাওয়া যাচ্ছে যেহেতু মিলে হবে। যেমন 64/64 মিলে আসছে।

 $\times$

(যদি 0 কোড হয়) bit এর 20 অপেক্ষা = 0

যদি 255 কোড হয় bit এর 20 অপেক্ষা = 255

Supernetting

❏ Subnetting could not completely solve address depletion problems in classful addressing because most ~~exam~~ organization did not want to share their granted blocks with others.

❏ Supernetting:

Supernetting is the process of combining several small networks into a large network.

⊗ A subnetmask for class C has more 1s than the default mask but the supernet mask has less 1s than the default mask for this class.

$$n_{\text{super}} = n - \log_2 c$$

n_{super} = length of supernetid in bits.

n = length of netid (for class C = 24)

c = Number of class C blocks that are combined.

- Supernetting brings in two new problems:
- 1) The number of blocks to be combined needs to be a power of 2. Which means, an organization needs seven blocks should be granted at least eight blocks. As a result, addresses are wasting.
 - 2) Subnetting and ~~sup~~ supernetting really complicated the routing of packets in the Internet.

classless Addressing

- What is ISP?
- ⇒ An ISP is an organization that provides internet access for individuals, small businesses and midsize organizations.
- Customers are connected via a dial-up modem, DSL or cable modem to the ISP.
- In classless addressing variable-length blocks are used that belongs to no classes.

▣ In classless addressing, we can have a block of $2^0, 2^1, 2^2, \dots, 2^{32}$ addresses.

▣ In classless addressing, a block is divided into two parts:

~~A suffix~~

i) prefix

ii) suffix

▣ The prefix plays the same role as the netid.

and the suffix plays the same role as the hostid.

▣ In classless addressing, the prefix defines the network and the suffix defines the host.

▣ In classless addressing the length of the prefix, n , depends on the size of the block. it can be 0, 1, 2, 3, ..., 32.

▣ In classless addressing, the value of n is referred to as prefix length. and the value of $(32-n)$ is referred to as suffix length.

Q The prefix length in classless addressing can be ~~1 to~~ 1 to 32.

Ex - 5.22

What is the prefix length and suffix length if the whole internet is considered as one single block with 4,294,967,296 addresses?

Solⁿ:

Prefix length, $n = 0$

Suffix length, $32 - n = 32 - 0$

$= 32$

Total address $N = 2^{32}$

$= 4,294,967,296$

Ex - 5.23

What is the prefix length and suffix length if the internet is divided into 4,294,967,296 blocks and one block has one single address?

Solⁿ:

Prefix length, $n = 32$

Suffix length, $(32 - n) = 32 - 32$

All 32 bits are needed to define $2^{32} = 4,294,967,$

296 blocks. The only address in each block is

Ex-5.26

The # number of addresses in a block is inversely related to the value of the prefix length, n .

A small n means larger block, and large n means small block.

☐ In classless addressing, the prefix length cannot be found if we are only given an address in the block.

The address can belong to a block with any prefix length.

☐ In classless addressing, we need to include the prefix length to each address if we need to find the block of the address.

☐ The slash notation is formally known as classless interdomain routing or CIDR notation.

☐ In classless addressing, we need to know one of the addresses in the block and the prefix length to define the block.

☐ Number of addresses in a block:

$$N = 2^{32-n}$$

N = Number of address

n = prefix length

☐ First address:

First address = (any address) AND (network mask)

Alternatively,

keep n leftmost bits as it is and set $(32-n)$ rightmost bits to 0's.

☐ Last address :

Last address = (any address) OR [NOT (network mask)]

Alternatively,

keep n leftmost bits of any address in the block and set $(32-n)$ rightmost bits to 1's.

"BLOCK ALLOCATION"

☐ The ultimate responsibility of block allocation is given to a global authority called Internet Corporation for Assigned Names and ~~Addresses~~ (ICANN) Numbers.

For the proper operation of the CIDR, three restrictions need to be applied to the allocated block.

i) The number of requested addresses, N , needs to be power of 2.

ii) The value of prefix length can be found from the number of addresses in the block.

$$N = 2^{32-n}$$

$$\Rightarrow n = \log_2(2^{32}/N)$$

$$\Rightarrow n = 32 - \log_2 N$$

iii) The beginning address of the block needs to be divisible by the number of the address in the block.

Ex - 5.30

An ISP has requested a block of 1000 addresses. The following block is granted:

a) Since, ~~1024~~¹⁰⁰⁰ is not a power of 2, 1024 addresses are granted. ($1024 = 2^{10}$)

b) The prefix length of the block is calculated as, $n = 32 - \log_2 1024$

c) The beginning address is chosen as 18.14.12.0 (which is divisible by 1024).

The granted block is $18.14.12.0/22$

The first address is $18.14.12.0/22$

The last address is $18.14.15.255/22$

Subnetting:

Three level of hierarchy can be created using subnetting.

Designing subnetworks:

- ⇒ The total number of address granted to the organization is N .
- ⇒ Prefix length is n .
- ⇒ Assigned number of addresses to each subnetwork is N_{sub} .
- ⇒ The prefix length of each subnetwork is n_{sub} .
- ⇒ The total number of subnetworks is S .

Rules to create subnetworks:

- 1) The number of address in each subnetwork should be a power of 2.
- 2) The prefix length for each subnetwork should be found using the following formula:

$$n_{sub} = n + \log_2 (N/N_{sub})$$

3) The starting address in each subnetwork should be divisible by the number of address in that subnetwork.

This can be achieved if we first assign address to the larger networks.

☐

Example: 5.32

An organization is granted a block 130.34.12.64/26. The organization needs four subnetworks, each with an equal number of hosts.

Design the subnetworks and find the information about each network.

⇒ Soln:

The number of address for the whole network

$$N = 2^{32-26} = 64$$

∴ The first address of the network is

$$130.34.12.64/26$$

∴ The last address of the network is

$$130.34.12.127/26$$

Now, we design the subnetworks:

- 1) We grant 16 addresses for each subnetwork to meet the first requirements

$64/16$ ~~is~~ is power of 2, [64 or 16...]

- 2) The subnetwork mask for each subnetwork is:

$$n_1 = n_2 = n_3 = n_4 = n + \log_2(64/16)$$

$$= 26 + \log_2 4$$

$$= 26 + 2\log_2 2$$

$$= 28$$

- 3) We grant 16 addresses to each subnet starting from the first available address.

